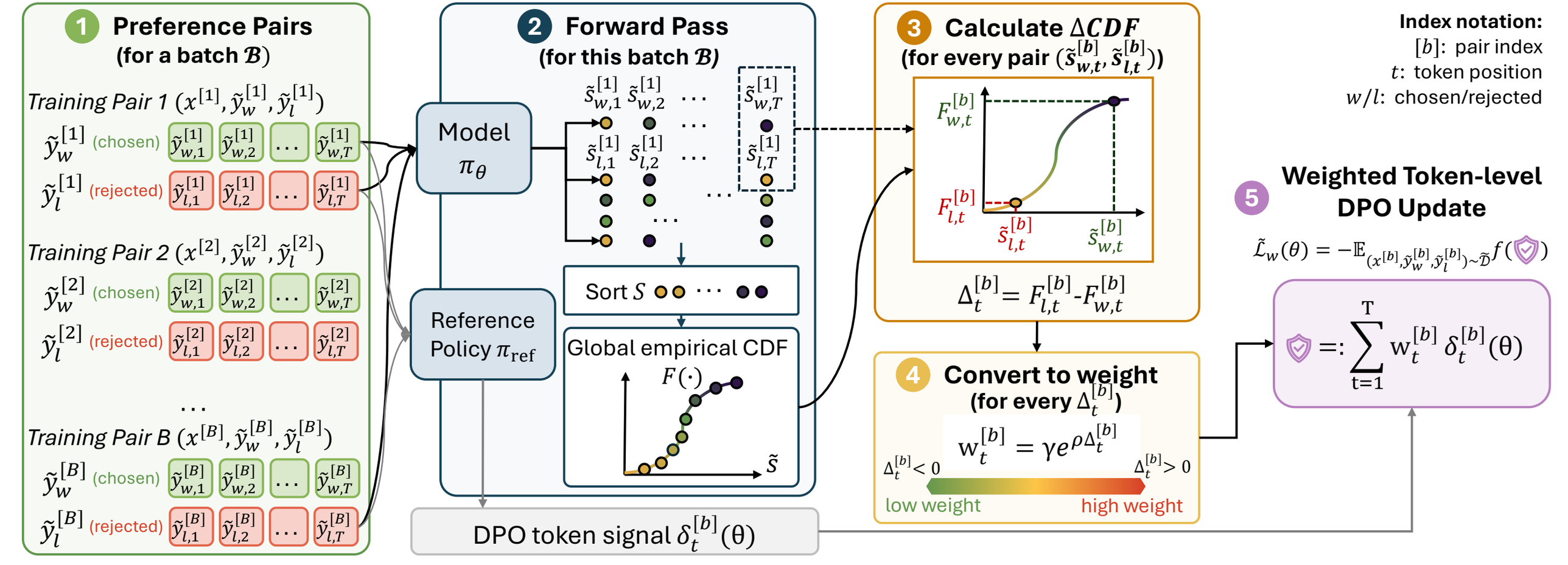

1 Preference Pairs
(for a batch $\mathcal{B}$)
Training Pair 1 $(x^{[1]}, \tilde{y}_w^{[1]}, \tilde{y}_l^{[1]})$
$\tilde{y}_w^{[1]}$ (chosen)
$\tilde{y}_l^{[1]}$ (rejected)
Training Pair 2 $(x^{[2]}, \tilde{y}_w^{[2]}, \tilde{y}_l^{[2]})$
$\tilde{y}_w^{[2]}$ (chosen)
$\tilde{y}_l^{[2]}$ (rejected)
Training Pair B $(x^{[B]}, \tilde{y}_w^{[B]}, \tilde{y}_l^{[B]})$
$\tilde{y}_w^{[B]}$ (chosen)
$\tilde{y}_l^{[B]}$ (rejected)
2 Forward Pass
(for this batch $\mathcal{B}$)
Model π_θ
Reference Policy π_{ref}
Sort S
Global empirical CDF
$F(\cdot)$
$\tilde{s}$
DPO token signal $\delta_t^{[b]}(\theta)$
3 Calculate ΔCDF
(for every pair $(\tilde{s}_{w,t}^{[b]}, \tilde{s}_{l,t}^{[b]})$)
$F_{w,t}^{[b]}$
$F_{l,t}^{[b]}$
$\tilde{s}_{l,t}^{[b]}$
$\tilde{s}_{w,t}^{[b]}$
$\Delta_t^{[b]} = F_{l,t}^{[b]} - F_{w,t}^{[b]}$
4 Convert to weight
(for every $\Delta_t^{[b]}$)
$w_t^{[b]} = \gamma e^{\rho \Delta_t^{[b]}}$
$\Delta_t^{[b]} < 0$
$\Delta_t^{[b]} > 0$
low weight
high weight
Index notation:
$[b]$: pair index
t: token position
w/l: chosen/rejected
5 Weighted Token-level DPO Update
$\tilde{\mathcal{L}}_w(\theta) = -\mathbb{E}_{(x^{[b]}, \tilde{y}_w^{[b]}, \tilde{y}_l^{[b]}) \sim \tilde{\mathcal{D}}} f(\cdot)$
$=: \sum_{t=1}^{T} w_t^{[b]} \delta_t^{[b]}(\theta)$